D-SCANet: A Plant-Level Leaf Area Estimation Method for In-Field Water Bamboo Using UAV RGB Remote Sensing Imagery

Yi Jiang, Yefan Zhang, Xiangying Xu, Qiancong Han, Yonglong Zhang, Zhiping Zhang, and Muhammad Jawaad Atif

***Abstract*—Leaves are vital organs governing plant photosynthesis, and leaf area (LA) quantification is crucial for assessing plant growth and yield. In recent years, unmanned aerial vehicle (UAV) remote sensing combined with machine vision approaches has been capable of estimating the canopy LA of plants by extracting leaf information from images. Nevertheless, for slender, dense-leaved plants, accurate plant-level LA estimation remains highly challenging since abundant leaves of varying sizes and overlapping distributions not only make annotating leaf boundaries labor-intensive during data processing, but also obscure leaf contours and shapes, thus preventing reliable area estimation during model prediction. To address these problems, a novel regression architecture was proposed for estimating the LA of water bamboo (*Zizania latifolia*), a typical dense-leaved aquatic vegetable. Without the need for leaf segmentation and annotation in images, the proposed end-to-end dynamic and structural context-attention network (D-SCANet) directly employs UAV-captured images, greatly reducing the data processing overhead for model training. It comprises a ConvNeXt backbone, an innovative serial structure for feature refinement, and a regression head. Specifically, the serial structure integrates two key components. One is a dynamic gated atrous spatial pyramid pooling (DG-ASPP) module, which adaptively aggregates multiscale contextual information via a data-driven gating mechanism. The other is a spatial coordinate attention with cross-dimensional interaction module (SCAM), which adopts a dual-branch design and incorporates a novel cross-dimensional interaction mechanism to perceive fine-grained spatial information. Results from comparative experiments demonstrate that D-SCANet achieves a mean absolute percentage error (MAPE) of 20.34% and a coefficient of determination (R^2) of 0.78, outperforming 13 state-of-the-art baseline models, whose MAPE values range from 28.83% to 22.62% and R^2 from 0.59 to 0.74. It indicates that the proposed model is capable of estimating crop canopy LA in complex field environments using automatically extracted features, eliminating the requirement for segmentation annotations. This approach offers a reference for phenotype identification of slender, dense-leaved crops under complex field conditions.**

***Index Terms*—Deep learning, leaf area (LA), regression, water bamboo phenotyping.**

I. Introduction

UNMANNED aerial vehicle (UAV) remote sensing in precision agriculture is widely recognized as a key strategy to enhance crop yield and resource use efficiency, as it optimizes field management via high-resolution data acquisition and analysis. Notably, this technology enables high-throughput and nondestructive monitoring of crops, providing essential data support for breeding, cultivation, and yield estimation. Among various crop phenotypes, leaf area (LA) is widely acknowledged as one of the primary determinants of crop productivity, as leaf size and shape profoundly influence key physiological processes such as photosynthesis, respiration, and transpiration [1]. Therefore, the precise measurement of LA has become a core approach for assessing crop growth status and predicting yield potential.

Traditional methods for LA measurement, such as the millimeter grid paper method [2] and planimeter scanning method [3], rely primarily on manual measurements. Although accurate, they are typically destructive, time-consuming, and labor-intensive, limiting their applicability for large-scale, high-frequency field monitoring [4]. Computer vision-based techniques have provided feasible solutions for nondestructive LA estimation. Previous image processing methods primarily relied on manually set color thresholding, edge detection, and morphological operations to segment leaf regions and calculate pixel area [5]. These methods exhibit limited robustness in complex field environments. In particular, they are highly susceptible to factors such as variations in illumination, shadows, background clutter (e.g., soil, water surfaces, and weeds), and severe mutual occlusion among leaves. Against this backdrop, deep learning method, particularly convolutional neural networks (CNNs), with their powerful capabilities for automated feature extraction and nonlinear modeling, has

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shown great potential in agricultural vision tasks [6]. However, direct estimation of LA values using deep learning approaches from RGB images taken in the field remains challenging, especially for crops with linear-lanceolate leaves such as water bamboo [7].

Water bamboo (*Z. latifolia*) is an aquatic vegetable, which is edible upon infection by the smut fungus (*Ustilago esculenta*) [8]. It is widely consumed throughout Asia and beyond for its crisp texture and high dietary fiber content. The plants of water bamboo can reach up to 2 m in height, with long, slender, and densely packed leaves, whose area is critical for photosynthesis and final yield. Normally growing in clusters on wet marshlands, the crop has strong tillering ability, with over 15 tillers at maximum and each tiller bearing more than eight leaves. Consequently, a single clump may have more than 100 leaves in total, which makes manual measurement of whole-plant LA extremely difficult in paddy fields, let alone the LA quantification of the entire field plot. So plot-scale estimation in agriculture is usually performed by sampling only a subset of plants or leaves. This, in turn, limits the availability of sufficient training data for machine vision-based methods.

In terms of remote sensing imagery, water bamboo fields exhibit distinct characteristics. On the one hand, dim light at the center of each clump makes inner leaves hard to distinguish in remote sensing images. On the other hand, green duckweeds commonly exist in paddy fields, whose color is highly similar to that of water bamboo leaves, making distinction challenging even with close-range remote sensing images. Moreover, dense crop foliage presents prominent multiscale variations, together with background clutter induced by water reflections, posing challenges for adaptive feature aggregation in conventional multiscale models. Meanwhile, intertwined leaf blades induce complex 2-D spatial occlusions. Conventional attention mechanisms struggle to effectively capture nonaxial structural relationships, including overlapping boundaries and junction points. The existing models suffer from inherent limitations in learning continuous LA values, including sensitivity to data noise and poor generalization under varying field conditions, leading to a shortage of methods directly applicable to LA estimation in water bamboo. Therefore, a convenient approach for quantifying LA at both the plant and field-plot levels is highly desirable.

This article proposes a new deep learning architecture, D-SCANet, to accurately estimate LA of water bamboo directly from RGB images taken by a UAV. Two specialized components are designed for adaptively capturing multiscale contextual information and enhancing perception capability regarding irregular edges and occluded regions. The main contributions of this study are summarized as follows.

1) An innovative water bamboo (*Z. latifolia*) LA dataset was constructed based on UAV-borne close-range remote sensing, where high-resolution RGB imagery was utilized to collect LA data of field-grown plants.
2) To address the size heterogeneity of water bamboo leaves, a novel dynamic gated atrous spatial pyramid pooling (DG-ASPP) module is proposed to realize adaptive and efficient aggregation of multiscale contextual information, thereby enhancing the accuracy of LA estimation.
3) A cross-dimensional interaction mechanism-based spatial coordinate attention with cross-dimensional interaction module (SCAM) is developed to improve the perception of leaf spatial structures and fine-grained phenotypic details.

The remainder of this article is organized as follows. Section II reviews related work. Section III details our proposed D-SCANet architecture. Section IV presents the experimental results. Section V discusses the implications and limitations of our findings. Finally, Section VI concludes this article.

II. Related Works

A. AI-Driven Remote Sensing and Plant Phenotyping

Compared with conventional remote sensing approaches including manual visual interpretation and shallow machine learning algorithms, artificial intelligence (AI)-driven remote sensing significantly enhances spatial mapping accuracy and realizes automated long-term dynamic monitoring. Satellite remote sensing assisted by AI supports multiple Earth observation applications, such as the monitoring of seagrass coverage, seawater color, and wetland boundaries [9], [10], [11]. UAV close-range remote sensing empowered by AI features low cost, fine spatial resolution, and flexible mobility compared with satellite remote sensing and exerts an increasingly important function within precision agriculture in recent years. For example, Cai et al. [12] proposed FastSegFormer, a lightweight real-time segmentation model for navel orange surface defect detection, which can be deployed on low-power onboard terminals to monitor orchard targets in real time. Tian et al. [13] adopted a multiscale CNN feature fusion algorithm for intelligent analysis of UAV remote sensing imagery, which effectively improves the recognition accuracy of ground objects at different spatial scales in farmland.

Recently, UAV-based plant phenotyping has gained widespread popularity for its capability to rapidly acquire high-throughput phenotypic traits of crops. Existing studies have employed deep-learning models to extract and analyze image features for plant morphological-structural phenotypes such as LA and biomass, physiological-biochemical phenotypes such as chlorophyll and nitrogen content, as well as stress-resistance phenotypes [14], [15]. Several studies have proposed few-shot and zero-shot monitoring strategies to alleviate the high cost associated with annotated data. For example, based on transfer learning, YOLO-SAM AgriScan fine-tunes the pretrained YOLOv11 with limited strawberry annotations, and employs pretrained SAM2 to realize zero-shot segmentation [16]. Currently, transfer learning approaches built upon visual deep learning models, vision-language models (VLMs), and large language models (LLMs) have gradually emerged to support zero-shot phenotypic recognition [15], [17]. Nevertheless, existing models exhibit competitive performance for qualitative phenotyping, while precise quantitative phenotypic regression remains challenging.

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B. Vision-Based Plant LA Estimation

Over the past few years, plant LA estimation from remote sensing imagery has emerged as a key topic in agricultural remote sensing. Studies have primarily focused on leaf segmentation in controlled or semi-controlled environments. These methods typically include color space conversion (e.g., RGB to HSV/Lab for enhanced color separation between crop leaves and background), adaptive image threshold approaches like Otsu's method, edge detection operators such as Canny or Sobel filters, and morphological operations including erosion and dilation for noise reduction [6], [18], [19]. Nevertheless, in complex field environments, challenges such as uneven illumination, shadows, soil/water reflections, and interleaf occlusion severely degrade the performance of these rule-based approaches, resulting in poor generalization and inability to meet the requirements of high-precision LA measurement.

Currently, a variety of computer vision-based methodologies have been developed for LA estimation in field crops, and these approaches are mainly divided into two types according to the data used: 2-D image-based and 3-D image-based [20]. Though 3-D image-based schemes generally capture the spatial structure of leaves well and yield superior results, the significant computational overhead and higher requirements on image acquisition devices limit their application [21]. The use of 2-D hyperspectral or multispectral images has been the mainstream approach for several years [22], [23]. However, the hyperspectral and multispectral cameras required for these methods are expensive, which renders the cost of image acquisition excessively high. In contrast, low-cost RGB images have received increasing attention in recent years to reduce reliance on high-cost equipment [24], [25], [26]. Nonetheless, there is still a gap at present in model performance between studies using low-cost RGB cameras or UAVs for image acquisition and those based on multispectral and hyperspectral images [25].

Recently, new solutions brought by improvements of model structures have significantly enhanced the capability to handle visual tasks in complex scenarios [27], [28], [29]. For dense prediction tasks such as LA estimation, there are currently two main solutions. One is a segmentation-preferred method where its backbone network mainly adopts convolutional operations or transformer-based architectures, such as U-Net or DeepLab. For instance, Ojo et al. [30] utilized a lightweight DeepLabv3 + network for high-precision semantic segmentation in their hydroponic lettuce phenotyping study. Based on the segmentation results, they further applied a multimodal regression network that incorporated RGB, depth, and geometric features to estimate key parameters, including LA and fresh biomass weight. This approach achieved excellent fitting performance, with a high R^2 value of 0.953. Da Silva et al. [31] introduced a dual-decoder structure into the DeepLabv3 + model to estimate bean's LA by segmenting the leaf and a reference marker, achieving favorable performance. Deng et al. [32] improved the UNet++ model by integrating shallow and deep features and performing multiscale feature fusion. Meanwhile, it introduces both the Outlooker attention mechanism and the Transformer attention mechanism to enhance the accuracy of leaf segmentation. Although these methods have achieved favorable results, most of the datasets they used are public. The workload of pixel-level annotation required for constructing datasets of dense leaves for specific crops is extremely heavy. For example, Simonyan and Zisserman et al. [35] reported that pixel-level annotation of maize seedling images takes 32.4 min per image, while bounding-box annotation only requires 2.5 min per image. Furthermore, He et al. [36] reported that pixel-level annotation of tomato plant images in complex scenes may require up to 5 h per image. This renders phenotypic recognition of specific crops still challenging.

The other solution is an annotation-free method that does not require prior segmentation of all pixels. This approach circumvents complex pixel-level segmentation processes and costly annotations and requires only image-level labels for training. Researchers typically adopt classical CNN architecture, such as VGG [35], ResNet [36], ConvNeXt [37], or EfficientNet [38], as feature extractors. Wittstruck et al. [39] demonstrated that a CNN model incorporating both RGB imagery and normalized digital surface model (nDSM) data achieved high accuracy for the estimation of winter wheat LA index (LAI), highlighting the performance gain from fused color-height features. Similarly, Ilniyaz et al. [1] conducted comprehensive comparisons of machine learning and ResNet-based deep learning models for UAV-derived LAI estimation in vineyard environments. These methods tend to result in relatively large errors when relying solely on RGB images and a single model, making it necessary to fuse multiple models and perform data augmentation. Zhang et al. [40] proposed a method for monitoring oat LAI using UAV multispectral images and a transfer learning framework, which achieved favorable results in LAI estimation across different growth stages of oats. Yang et al. [41] captured maize leaf images with a smartphone and developed an improved CNN for maize LA estimation, with the model's R^2 exceeding 0.8. To the best of the authors' knowledge, current research on field-scale LA estimation for dense crops such as water bamboo is insufficient, and studies on direct LA estimation using unlabeled RGB images from UAVs still require further exploration.

C. Dense and Small Object Extraction in Computer Vision

1) Multiscale Feature Fusion: In the calculation of dense LA, due to the varying sizes of leaves, it is difficult to accurately extract leaf regions using conventional convolution operations in CNN, which has a unified kernel size. Therefore, it is necessary to extract features of leaves with different sizes and fuse multiscale features accordingly. Feature pyramid networks (FPNs) are capable of fusing feature maps from different stages of a network backbone, thus emerging as a powerful tool for multiscale feature fusion. It utilizes a top-down pathway with lateral connections to effectively combine deep, semantic features with shallow, high-resolution features, which has been widely adopted and continuously refined in multiscale vision tasks. For instance, Hu et al. [42] utilized FPN in their OrchidNet to fuse multilevel features for precise orchid classification. More recently, with the rise of Vision Transformer (ViT), researchers have explored integrating multiscale principles into transformer architectures.

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Xia et al. [43], for example, proposed ViT-CoMer, which introduces a convolutional multiscale feature interaction module to enable a pure ViT model to effectively handle dense prediction tasks that require robust multiscale representations.

Atrous spatial pyramid pooling (ASPP) [44] represents another highly influential solution for fusing multiple features, which is popularized in the DeepLab series models. ASPP employs parallel atrous convolutions with distinct dilation rates to capture multiscale context without increasing computational costs. This method has been widely adopted in dense estimation tasks. For example, Diao et al. [45] integrated ASPP into U-Net architectures to enhance maize crop row segmentation in complex field conditions. Similarly, in medical imaging, Shahin et al. [46] also incorporated an ASPP module into their network to capture informative multiscale context for brain tumor classification. However, these classical fusion mechanisms are typically static. The standard ASPP treats all scale branches equally without the flexibility to adaptively adjust based on input content, while its channel concatenation approach incurs high computational costs.

Recent works have aimed to improve ASPP by making it more dynamic and efficient. For instance, Qiu et al. [47] proposed attentive ASPP (A2SPP), which integrates a heavy 3-D attention module (CIEA) into each parallel branch to enable spatial-pixel-level scale selection. Moreover, performing dense pixelwise weighting locally renders the model highly susceptible to high-frequency noise (such as specular water reflections). To address the parameter overhead, Wang et al. [48] introduced window attention-ASPP (WA-ASPP) by replacing dilated convolutions with window-based self-attention. However, this method relies on static, predefined window partitioning, which lacks the flexibility to adaptively adjust its attention boundaries to match the highly irregular, overlapping contours of plant leaves, while simultaneously discarding the local inductive bias of convolutional layers that is critical for capturing fine, high-frequency boundary textures. Similarly, alternative methods utilizing deformable attention or dynamic convolutions introduce an excessive gradient optimization burden in regression tasks by attempting to learn continuous spatial offsets or kernel weights under weak global supervision. Consequently, developing a multiscale feature fusion mechanism that simultaneously achieves robust, content-adaptive scaling and low computational complexity remains an active and unresolved challenge in complex plant phenotyping.

2) Attention Mechanisms: The attention mechanism, designed to emulate the human visual system by enabling models to focus on the most critical information, has become a pivotal technique for enhancing model performance. Nowadays, the attention mechanism has evolved into several technical branches, including channel, spatial, and hybrid attention. The squeeze-and-excitation network (SE-Net) [49] represents a milestone in channel attention, which recalibrates feature responses by learning interchannel dependencies. However, SE-Net encodes channel relationships through global pooling, losing spatial positional information, which is a significant limitation for tasks requiring irregular shape localization. Correspondingly, spatial attention [50] focuses on and enhances key spatial regions while suppressing interference from irrelevant background regions, thereby improving the model's ability to capture spatial context. As a hybrid attention mechanism, coordinate attention (CoordAtt) [51] embeds positional information into channel attention by decomposing the 2-D feature map into two separate 1-D feature encoding processes, thereby capturing direction-aware long-range dependencies while maintaining a lightweight architecture.



Fig. 1. DJI MINI 2 SE.

However, standard CoordAtt relies solely on average pooling to encode coordinate information, which primarily captures general feature statistics while potentially overlooking the most salient features. Notably, Gupta and Tripathi [52] proposed an enhanced module called dual coordinate attention feature extraction (DCAFE) in their research of medicinal flower identification to overcome this limitation. By integrating both average and max pooling operations in parallel, DCAFE simultaneously captures broad feature distributions and high-discriminability details of floral characteristics. But it still relies on an assumption that 2-D spatial attention can be decomposed into two independent 1-D vector processes, which is likely to sacrifice the complex interrelationships inherent in the 2-D spatial domain.

III. Proposed Methods

A. Data Acquisition

Data acquisition was conducted in the fall of 2023 and 2024 in the water bamboo experimental fields of the vegetable base at Yangzhou University and in Shatou Town, Yangzhou, Jiangsu, China. Five experimental plots were established in each year and demarcated using separation panels. Each plot received a different level of nitrogen fertilizer treatment.

A DJI MINI 2 SE (see Fig. 1) drone equipped with an RGB camera was used for image acquisition. Flight altitude was maintained at 10 m above ground level to ensure sufficient spatial resolution for clearly distinguishing water bamboo leaves in the captured images.

Image capturing was performed between 12:00 P.M. and 5:00 P.M. under clear or lightly clouded weather conditions to minimize variations in illumination and shadow effects. For each plot, 5–6 independent flight missions were conducted. Each flight employed different camera angles (90°, 85°, 80°, and 75°). This multiangle acquisition strategy aimed to enrich training data with diverse practical perspectives and thus improve the model's robustness to viewpoint variations. The relevant parameter information is shown in Table I.

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TABLE I
UAV FLIGHT MISSION PARAMETERS

Parameter	Specification
Model	DJI Mini 2 SE
Weight	< 249 g
Flight Time	31 min
Flight Mode	Normal Mode
Camera Model	Integrated Wide-Angle Camera
Flight Mode	1/2.3-inch CMOS
Image Resolution	4000 x 2250
Flight Altitude	10m
Data acquisition time period	12:00PM-5:00PM
Gimbal Pitch Angle	90 °, 85° , 80° , 75°, etc
Shooting environment	Clear or lightly cloudy

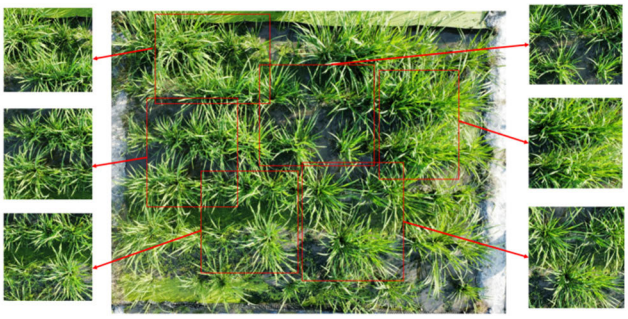


Fig. 2. Original image and subimages.

TABLE II
DETAILED INFORMATION OF THE DATASETS

Date of image collection	Original images	After image cropping
2024-9-26	40	539
2024-10-09	36	445
2024-10-25	20	269
2024-11-13	47	352
2023-09-10	5	
2023-09-26	5	
2023-10-10	5	
Total	158	1620

B. Data Preprocessing

Following drone-based aerial imaging, regions of interest (ROIs) containing target water bamboo were extracted from the original images, which included nontarget areas such as field ridges and background vegetation. To increase the number of samples and reduce the complexity of images, the original images were randomly cropped into subimages containing single or a few plants (see Fig. 2). After image cropping, the total number of samples reaches 1605. The detailed information of the datasets is shown in Table II.

Considering the substantial workload associated with direct physical measurements for all samples, a hierarchical labeling strategy integrating direct measurement and expert estimation was adopted. Specifically, several plants per plot were selected for physical measurement, and based on these precisely measured plants (173 samples) as references, the LA of plants in the remaining cropped images was estimated by three researchers. The final ground-truth label for each estimated sample was determined by taking the mathematical average (mean value) of these three independent estimations. These manual estimations served as training labels for the model. Although the estimation method is relatively coarse, it takes an average of 5 min per sample. In contrast, annotation of a segmented image takes a minimum of 20 min, with finer annotation requiring more time. This indicates that the estimation method drastically cuts down the dataset processing time.

To comply with the fixed input size specification of deep learning models while preserving the original aspect ratio of the plants, all subimages underwent a standardized two-step preprocessing. First, each image was proportionally resized according to a fixed scaling factor (0.4245), which was determined by calculating the minimum scaling ratio required for the largest image in the dataset, specifically, the smaller of the ratios between the target dimensions and the original dimensions of the largest image in both directions. This resizing operation, crucial for maintaining image integrity, utilized Lanczos resampling for superior image quality. Subsequently, this scaled image was padded and centered onto a 512×512 white background. This padding strategy ensures that all inputs to the model are of a uniform size without distorting the plant's morphology.

C. Dynamic and Structured Context-Attention Network

To accurately estimate LA of water bamboo from complex field environments, we designed and proposed a novel deep neural network architecture named dynamic and structural context-attention network (D-SCANet), as shown in Fig. 3. The overall architecture of D-SCANet systematically comprises three main components: a hierarchical feature extraction backbone (consisting of Stages 0–4), a feature enhancement stage (DG-ASPP and SCAM), and a regression head. Instead of employing generic attention modules, we developed two specialized, sequentially connected enhancement modules that cooperatively tackle the key challenges in LA estimation, including adaptive aggregation of multiscale contextual information and precise perception of spatial features in leaf images. First, we introduced the DG-ASPP module. Distinct from standard ASPP, DG-ASPP employs a data-driven gating mechanism to dynamically assign weights to receptive fields of different scales, followed by efficient weighted summation for feature fusion, enabling the model to intelligently aggregate multiscale contextual information based on image content. Subsequently, we designed the SCAM to process the output from DG-ASPP. SCAM captures both global statistics and salient details through parallel mean and maximum branches, with its core innovation being a cross-dimensional interaction mechanism that explicitly models second-order relationships between horizontal and vertical 2-D attention maps that significantly enhance the model's perception of spatial features in leaf images, particularly irregular edges and junction points.

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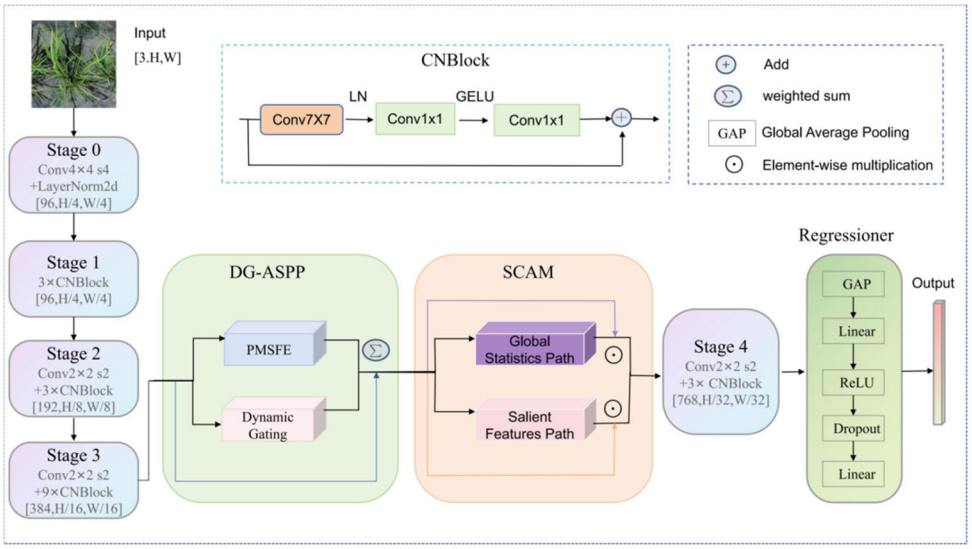


Fig. 3. Overall architecture of D-SCANet.

1) Feature Extraction Backbone: The backbone network of D-SCANet is responsible for extracting hierarchical deep features from input images of water bamboo. ConvNeXt network is used as this backbone here. It is structured into an initial stem layer (Stage 0) and four subsequent stages (Stages 1–4). The stem layer performs preliminary downsampling and feature projection. Following this, each subsequent stage, typically beginning with a downsampling convolution, leverages a varying number of CNBlocks for feature learning and transformation. This multistage design ensures the progressive reduction of spatial resolution in feature maps, simultaneously expanding the receptive field and enriching the channel representations. Notably, while Stages 0–3 form a sequential pipeline, Stage 4 uniquely processes features after integration with enhanced modules, continuing feature refinement and downsampling.

The CNBlock (see Fig. 3) employs a large-kernel depthwise convolution (7 × 7 Conv) to capture local region information, stabilized via layer normalization (LN). Subsequently, two bottleneck-style 1 × 1 convolutions coupled with GELU activation functions perform interchannel feature transformation and nonlinear mapping.

2) DG-ASPP Module: The task of this study is to accurately estimate the total LA of water bamboo, which comprises individual leaves of different sizes, from images captured in complex field environments. We propose a DG-ASPP module specifically designed for this task. Its core design is to dynamically assign weights to multiscale features in a data-driven and adaptive manner, thereby enabling adaptive and efficient feature fusion for LA estimation in complex field environments. The overall architecture of DG-ASPP (see Fig. 4) consists of four components: parallel multiscale feature extraction (PMSFE), dynamic gating mechanism, weighted feature fusion, and residual projection connection.

PMSFE is designed to comprehensively capture leaf information at different hierarchical levels. Normalization is applied to the input feature map $X \in R^{N\times C\times H\times W}$ and then a K

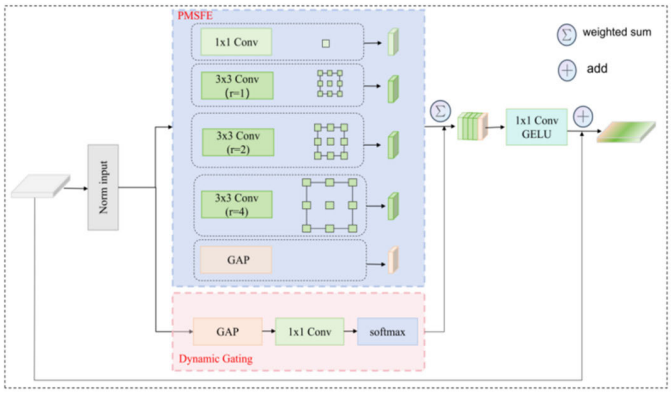


Fig. 4. Architecture of the DG-ASPP module. GAP denotes global average pooling, and PMSFE represents the parallel multiscale feature extraction.

parallel branches model is designed, where the 1 × 1 convolution branch preserves high-frequency details, corresponding to leaf edges and textures; the multiple dilated convolution branches, employing 3 × 3 convolutions with different dilation rates r ∈ {1, 2, 4} acquire receptive fields of varying sizes, corresponding to contexts like single leaves or leaf clusters; the global average pooling branch provides the model with a highly generalized representation of the overall image content by performing pooling operations across the entire feature map. All branches are designed with an output channel number identical to the input channel number C to facilitate efficient subsequent fusion. The output of the *k*th branch is denoted as $F_k \in R^{N\times C\times H\times W}$.

The dynamic gating mechanism is key to achieving adaptive fusion. A lightweight gating network is designed to generate attention weights for each branch of the feature map. The spatial dimensions of the input feature maps are first compressed using global average pooling to generate a global context vector $z \in R^{N\times C\times 1\times 1}$, which summarizes information from the entire feature map.

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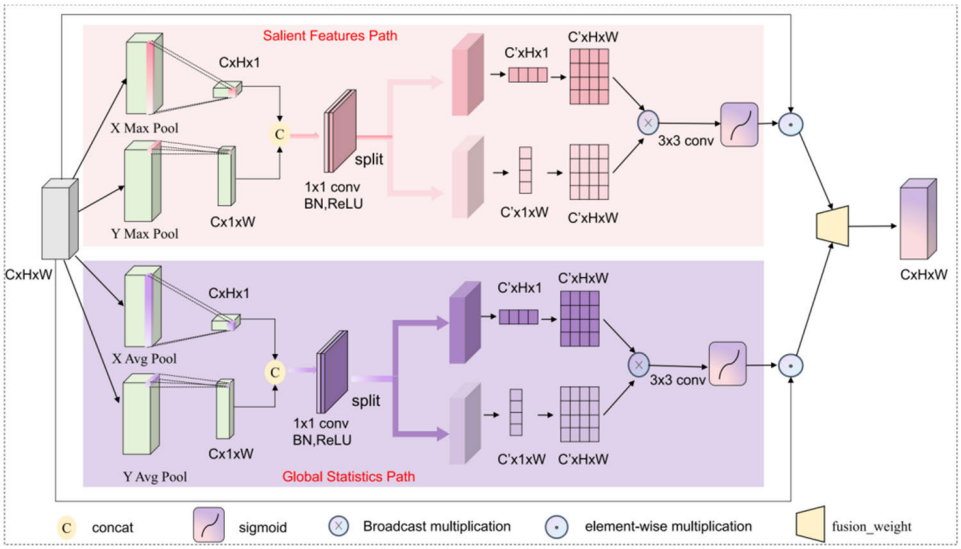


Fig. 5. Structure of SCAM module.

Then, a 1×1 convolutional layer models dependencies between branches. This layer maps the global vector z with C channels to a score vector s with K channels (total number of branches, K = 5). Each element in s represents the importance score of the corresponding branch

$$s = \mathrm{Conv}_{1\times1}^{C->K}(z). \tag{1}$$

Finally, a Softmax function is applied along the channel dimension (dim = 1) to normalize the scores s, producing a set of dynamic weights $w \in R^{N\times K\times1\times1}$ that sum to 1

$$w_k = \frac{\exp(s_k)}{\sum_{j=1}^{K}\exp(s_j)} \tag{2}$$

where w_k is the dynamic weight for the *k*th branch. This weight is generated dynamically based on the input content, enabling the model to intelligently adjust its reliance on information from different scales for different images or regions. We abandon the computationally expensive channel concatenation operation traditionally used in ASPP in favor of a weighted summation approach for fusing multiscale features. Using dynamic weights w, the fused feature F_{fused} is computed as follows:

$$F_{fused} = \sum_{k=1}^{K} w_k \odot F_k \tag{3}$$

where $\odot$ denotes the elementwise multiplication after broadcasting. This weighted summation strategy enables an adaptive fusion of features by dynamically weighting each scale, which avoids increasing the channel dimension.

To enable DG-ASPP to serve as a stable plug-and-play module while facilitating efficient gradient flow in deep networks, a residual connection is introduced. After being processed through a 1×1 convolution and GELU activation, the fused feature is added to the module's original input.

3) SCAM: To overcome the limitations of singular information encoding and a dimensional independence assumption, we design a novel SCAM to precisely locate and enhance the feature regions in the image (see Fig. 5). A parallel process of both mean and max pooling information is proposed, and a cross-dimensional interaction mechanism is introduced to explicitly model 2-D spatial dependencies.

The SCAM module is designed to apply refined position-aware weights to the feature map. Its workflow can be decomposed into three key steps.

1) *Dual-Branch Coordinate Information Encoding:* Given an input feature map $X_{in} \in R^{N\times C\times H\times W}$, we first perform feature encoding along the horizontal (H) and vertical (W) directions in two parallel branches and use average pooling to aggregate features for each row and column, capturing the average feature response of regions. This is crucial for identifying the smooth main body areas of leaves. In parallel, use max pooling to encode coordinate information. This captures the most salient feature responses on each row and column, which is vital for locating high-frequency details such as leaf edges, highlight points, and fine textures.
2) *Cross-Dimensional Interaction Attention Generation:* It abandons the assumption of dimensional independence and combines the horizontal and vertical features. Taking the mean branch as an example to elucidate the process: first, the horizontally and vertically oriented feature vectors $z_{h,avg}$ and $z_{w,avg}$ obtained from the mean branch are concatenated and then downsampled via a shared 11 convolutional layer, yielding intermediate features $f_{h,mean} \in R^{C'\times H\times1}$ and $f_{w,mean} \in R^{C'\times1\times W}$. Then, we compute their outer product through broadcasted multiplication, thereby generating a 2-D interaction map $M_{mean} \in R^{C'\times H\times W}$. Thus, each element $M_{mean} \in (i, j)$ within this map simultaneously encodes aggregated information from both the *i*th row and *j*th column of the original feature map. This enables the model to explicitly model second-order dependencies between the two dimensions. Subsequently, a 3×3 convolutional layer is adpoted, which is to learn the local spatial patterns within the interaction map and generate the

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full 2-D attention map $A_{mean} \in R^{C\times H\times W}$. Finally, a Sigmoid function is applied to normalize its values to the range (0, 1)

$$M_{mean} = f_{h,mean} \otimes f_{w,mean} \tag{4}$$

$$A_{mean} = \sigma\left(Conv_{3\times 3}\left(M_{mean}\right)\right) \tag{5}$$

where σ denotes the Sigmoid function. The max branch follows an identical process, generating its own dedicated 2-D attention map A_{max} through its independent network layers.

3) *Learnable Weighted Fusion:* We adaptively fuse the two sets of refined feature maps. Both 2-D attention maps are applied to the original input feature map X

$$Y_{mean} = X_{in} \odot A_{mean} \tag{6}$$

$$Y_{max} = X_{in} \odot A_{max} \tag{7}$$

where $\odot$ denotes the elementwise multiplication.

To intelligently combine these two enhanced representations, we introduce a learnable scalar parameter α, which is transformed into a fusion weight $w \in (0, 1)$ using the Sigmoid function

$$w = \sigma(\alpha). \tag{8}$$

The final output of the module Y_{SCAM} is the weighted sum of these enhanced feature maps

$$Y_{SCAM} = w \cdot Y_{mean} + (1 - w) \cdot Y_{max}. \tag{9}$$

IV. Experimental Results

A. Experimental Setup

All experiments were conducted under a unified hardware and software environment. The experimental platform consisted of a computing card equipped with a 16-core Intel[1] Xeon[1] Platinum 8474C CPU, 80 GB of system memory, and one NVIDIA RTX 4090D GPU with 24 GB of VRAM. All training and inference tasks were performed on the GPU. The operating system was Ubuntu 20.04. We used PyTorch 2.0.0 as the deep learning framework, Python 3.8 as the programming language, with GPU acceleration supported by CUDA 11.8.

We trained the proposed D-SCANet model end-to-end, employing AdamW as the optimizer. We adopted ConvNeXt-Tiny as our feature extraction backbone, initializing its parameters with pretrained weights on the ImageNet-1K dataset to provide a rich prior for multilevel visual feature representations. Compared with conventional Adam, AdamW provides more effective weight decay regularization, enhancing model generalization capability, with its weight decay parameter set to $1e^{-5}$ for superior regularization efficacy. To fully leverage pretrained knowledge from the backbone network, we implemented a differential learning rate strategy. Specifically, model parameters were divided into two groups: Backbone and Head. For the pretrained backbone with robust generic feature extraction capabilities, we assigned an extremely low learning rate of $1e^{-6}$ for cautious fine-tuning. Conversely, for the DG-ASPP and SCAM modules along with the final MLP regression head, we applied a comparatively higher learning rate of $1e^{-3}$ to accelerate adaptation to the new task. Learning rate adjustment followed a combined scheduler incorporating warm-up and cosine annealing (Fig. S1 of the Supplementary Material).

During the initial five epochs, the learning rate linearly increased from $1e^{-7}$ to the target values ($1e^{-3}$), preventing early training instability; subsequently, it smoothly decayed along a cosine annealing curve to a near-zero minimum throughout the remaining epochs. Given the regression task of estimating continuous LA values, we adopted mean squared error (mse) loss as the training objective. To eliminate the risk of label leakage and sample similarity bias, for the 2024 dataset, we designated four complete subplots exclusively for the training set, while reserving the remaining single subplot as the independent testing set. This guarantees that there is zero sample overlap or shared source imagery between the training and testing phases. To enhance the model's robustness against field variability and mitigate overfitting on limited training data, we implemented online data augmentation during training. The model was trained for 200 epochs with a batch size of 8, with all training and evaluation conducted within the PyTorch framework.

TABLE III
Estimation Results of Our Proposed D-SCANet on Test Samples

Case	Original image	Description	Estimated values	Manually labeled values	Percent error
(a)		Plant leaves are relatively sparse	5773.08	5845.83	1.2%
(b)		The background is clear	3588.26	3497.22	2.6%
(c)		Plant leaves are rela- tively dense	12102.85	13188.62	8.2%
(d)		The background water is covered with duck- weed	15075.56	17099.12	11.8%

B. Estimation Results

To more intuitively evaluate the practical performance and robustness of our model, we selected representative samples from the test set, covering variations in leaf density, lighting conditions, and background colors, to verify the model's effectiveness (see Table III). When plant leaves are relatively sparse and the background is clear in images, the model's prediction error remains within a low range [see Table III, Case (a) and (b)]. When the background water is covered with a large amount of duckweed, which has a similar color to the leaves and constitutes significant background noise, there is an approximate 11.8% error between the model's estimated values and the true values [Table III, Case (d)].

To assess the model's overall performance, we plotted a scatter plot of true LA estimated values for all test samples, as shown in Fig. 6(a). The plot intuitively shows that the data

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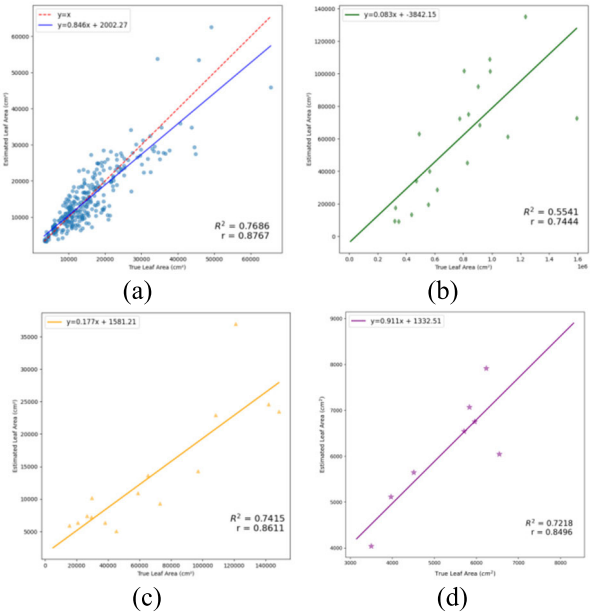


Fig. 6. Scatter plots of measured LA versus LA estimated by the proposed D-SCANet on the test set and two distinct experimental plots in 2023 and 2024. (a) Scatter plot of true leaf area versus estimated LA on the test set. (b) Scatter plot of true LA versus estimated LA at five subplots (2024). (c) Scatter plot of true LA versus estimated LA at five subplots (2023). (d) Estimated results from manual measurements on the test set.

points are tightly clustered around the perfect prediction line (y = x), exhibiting a strong positive linear relationship with the Pearson correlation coefficient (r) of 0.8767. To directly address the crucial requirement for cross-site validation, Fig. 6(b) and (c) presents the estimation results aggregated at the subplot level, with the 2023 dataset [see Fig. 6(c)] being collected from an entirely independent test site characterized by distinct environmental and soil conditions. From the figure, we can observe that the model has a good linear fit (R^2 = 0.5541 and r = 0.7444) in 2024 subplots and demonstrates an exceptionally strong, robust correlation at the independent 2023 site (R^2 = 0.7415 and r = 0.8611). Furthermore, on the subset of test samples verified by direct physical measurements [see Fig. 6(d)], which consists of eight ground-truth measurements, each corresponding to the average estimation from six multiangle images, D-SCANet demonstrated a good linear agreement with ground truth (R^2 = 0.7218, r = 0.8496). These findings collectively substantiate the cross-site generalization capability of D-SCANet, proving its practical applicability beyond the training domain.

To comprehensively evaluate the performance of our proposed D-SCANet architecture, we conducted comparative experiments with a series of classical and modern backbone networks widely used in computer vision, spanning classic CNNs, modern lightweight CNNs, standard ViTs, and recent hybrid architectures. All models were trained on our self-constructed water bamboo dataset using the same optimized fine-tuning strategy and evaluated on an independent test set.

The experimental results are summarized in Table IV. It reveals that standard backbone networks and recently proposed advanced architectures exhibit varying degrees of performance bottlenecks when directly applied to this LA regression task. Among the comparison models, DenseNet121 and the recently proposed OverLoCK emerge as the strongest competitors. DenseNet121 achieves a notable R^2 of 0.74 and a mean absolute percentage error (MAPE) of 22.64%, benefiting from its dense feature reuse mechanism. OverLoCK, which leverages local-to-global attention, achieves an R^2 of 0.72 and a MAPE of 22.99%. However, these comparison models fail to yield optimal predictive accuracy, indicating that standard representations remain constrained when resolving the multiscale variations characteristic of dense agricultural canopies.

In contrast, our proposed D-SCANet achieves superior performance across all accuracy-related metrics. Specifically, D-SCANet records the lowest MAPE (20.34%), which is 2.30% and 2.28% lower than those of the strongest CNN-based baseline, DenseNet121 (22.64%), and the strongest Transformer-based baseline, CAS-ViT (22.62%), respectively. More importantly, D-SCANet attains the highest R^2 (0.78) and Pearson Correlation coefficient (0.88), indicating its exceptional capacity to explain data variance and capture the true linear trends of LA. This substantial improvement quantitatively validates the efficacy of our cascaded design.

The robustness of D-SCANet is further substantiated by its error metrics. Our model achieves the lowest MAE and RMSE among all comparison models. It is noteworthy that while DenseNet121 exhibits a relatively close MAPE to our model, its RMSE is significantly higher.

Table IV also presents a tradeoff between model performance and computational efficiency. From the perspective of parameter efficiency, lightweight networks like

MobileNet (3.46M) and EfficientNet (4.66M) offer extremely compact footprints but suffer from suboptimal accuracy (MAPE of 25.81% and 23.35%, respectively). Conversely, modern heavy-weight models like OverLoCK (95.48M) and DeiT (86.04M) incur prohibitive parameter costs and severe latency bottlenecks. For instance, OverLoCK recorded an inference latency of 29.81 ms and only 33.55 frames/s without achieving proportional performance gains.

D-SCANet successfully achieves a favorable tradeoff between complexity and speed with 48.20M parameters and 48.29G FLOPs. Despite its moderate parameter count, D-SCANet achieves an exceptionally low inference latency of 3.14 ms and a high processing speed of 318.98 frames/s on the test hardware, outperforming all other heavy-weight Transformer and hybrid models. For reference, it runs approximately 9.5 times faster than OverLoCK and 3.2 times faster than DeiT. It is attributed to our nonredundant gating and structural coordinate attention designs, which bypass the quadratic computational overhead associated with standard self-attention mechanisms. Consequently, D-SCANet demonstrates practical viability for deployment on resource-constrained agricultural edge devices where real-time inference is paramount. Furthermore, an uncertainty analysis of the model's prediction errors confirms that the majority of test samples are clustered within a 15% error margin, with the distribution centered near the zero-error threshold, demonstrating consistent estimation stability across typical field canopies. The complete frequency

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TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT NETWORKS ON TEST SET. D-SCANET (OURS) IS BASED ON THE CONVNEXT-TINY BACKBONE

model	MAPE (%)	R^2	Pearson Correlation	params(M)	FLOPs	Inference time(ms)	FPS	MAE (cm^2)	RMSE(cm^2)
ResNet50	28.83	0.70	0.83	24.56	170.83	6.19	161.58	3983.27	5643.69
DenseNet121	22.64	0.74	0.86	7.47	118.42	4.95	202.11	3533.10	5679.97
MobileNet	25.81	0.65	0.80	**3.46**	8.90	**1.94**	**515.81**	3886.65	5558.95
VGG	23.47	0.68	0.82	14.98	641.99	4.65	215.07	3561.74	5145.00
EfficientNet	23.35	0.73	0.85	4.66	16.06	2.30	435.66	3529.10	5095.83
ConvNeXt-V2 [54]	25.91	0.59	0.77	88.2	80.37	10.85	92.13	4083.15	6021.87
OverLoCK [55]	22.99	0.72	0.85	95.48	92.46	29.81	33.55	3768.17	5457.66
MobileMamba [56]	26.35	0.66	0.81	16.90	33.68	3.02	331.30	3860.02	5324.94
ViT	24.80	0.63	0.80	86.82	5.29	7.39	135.32	3829.09	5675.78
Swin-T	27.00	0.61	0.78	27.91	7.7	3.91	255.43	4049.26	5893.89
PVT [57]	24.85	0.67	0.82	25.10	22.76	4.40	227.30	3748.87	5219.03
DeiT	23.94	0.67	0.82	86.04	87.74	10.07	99.35	3650.91	5319.76
CAS-ViT [58]	22.62	0.66	0.81	11.85	9.85	3.74	267.14	3556.41	5316.59
D-SCANet	**20.34**	**0.78**	**0.88**	48.20	48.29	3.14	318.98	**3089.81**	**4277.50**

TABLE V
THIS IS A SAMPLE OF A TABLE TITLE ABLATION STUDY ON THE EFFECTIVENESS OF THE DG-ASPP AND SCAM MODULES

DG-ASPP	SCAM	MAPE(%)	R^2	Pearson Correlation	params(M)	FLOPs	Inference time(ms)	FPS	MAE(cm^2)	RMSE(cm^2)
×	×	23.49	0.66	0.81	**43.68**	**12.57**	5.66	176.72	3972.11	6025.17
×	√	21.26	0.77	0.87	43.77	13.26	**2.75**	**363.76**	3154.96	4295.86
√	×	21.17	0.75	0.87	48.11	47.60	3.00	333.61	3186.75	4499.12
√	√	**20.34**	**0.78**	**0.88**	48.20	48.29	3.14	318.98	**3089.81**	**4277.50**

TABLE VI
COMPARATIVE ABLATION STUDY OF THE PROPOSED DG-ASPP AND SCAM MODULES AGAINST STANDARD ASPP AND COORDINATE ATTENTION BASELINES

Modules	MAPE(%)	R^2	Pearson Correlation
ASPP	28.61	0.62	0.79
Coordinate Attention	21.50	0.72	0.84
DG-ASPP+SCAM	**20.34**	**0.78**	**0.88**

distribution and kernel density estimate plot are detailed in Fig. S2.

To verify whether the observed improvements in accuracy are statistically significant, we performed two-tailed paired t-tests on the absolute percentage errors (APEs) of all test samples (Table S1). The statistical analysis consistently yields negative t-statistics across all 13 compared configurations. This proves that the prediction errors of D-SCANet are lower than those of the comparative networks. Furthermore, the calculated p-values are small. The null hypothesis is rejected at the 5% significance level or better for all comparisons. This formal testing confirms that the accuracy improvements of D-SCANet are systematically derived from the synergistic integration of the DG-ASPP and SCAM modules, rather than random sampling variations in the test dataset.

C. Ablation Study

To precisely quantify the individual contributions of the DG-ASPP and SCAM modules, we conducted a comprehensive ablation study (see Table V). Using ConvNeXt-Tiny as the backbone network, four different model configurations are compared. The baseline model, consisting solely of the ConvNeXt-Tiny backbone and an MLP regression head, established a performance benchmark with a MAPE of 23.49%. When the SCAM module was introduced independently, the model's performance improved notably, with the MAPE decreasing to 21.26%. This represents an error reduction of approximately 2.23%, providing clear evidence for the effectiveness of the SCAM module.

DG-ASPP also delivers a significant performance improvement, outperforming the configuration with only the SCAM module by reducing the MAPE to 21.17% and corresponds to a 2.32% reduction in error relative to the baseline. Simultaneous integration of both DG-ASPP and SCAM modules in our final D-SCANet architecture yields the lowest MAPE of 20.34%, representing the best performance among all configurations.

Specifically, adding SCAM to the model with the DG-ASPP module further reduces the MAPE by an additional 0.83%. Similarly, incorporating DG-ASPP into the SCAM-enhanced model leads to an extra 0.92% reduction in MAPE.

To further validate the specific architectural advantages of DG-ASPP and SCAM over their foundational predecessors, we conducted a component-replacement ablation study on the test set. As presented in Table VI, our proposed DG-ASPP and SCAM combination achieves the best performance across all metrics. Compared with the standard ASPP baseline, our architecture reduces the relative error by 8.27% while increasing the R^2 by 0.16, confirming that the dynamic soft-routing gating mechanism outperforms conventional static concatenation in multiscale feature aggregation. Furthermore,

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our model also consistently outperforms the standard coordinate attention baseline, verifying that the dual-branch pooling and 2-D cross-dimensional spatial interaction in SCAM effectively capture complex leaf intersection and overlapping boundaries that separable 1-D attention mechanisms cannot resolve.

Additional ablation studies were conducted to investigate the impact of varying module placement stages, dilation rate combinations, and architectural integration strategies (serial versus parallel). Detailed quantitative comparison presented in Tables S2–S4 of the Supplementary Material further validate the rationality and effectiveness of our proposed module design.

D. Qualitative Analysis and Interpretability

To further investigate the specific performance differences and potential systemic biases of the models beyond aggregate metrics, we plotted scatter plots of true values versus estimated values for each model on the test set (Fig. S3). The scatter plots illustrate that all models capture the positive correlation between true and estimated LA. Our proposed D-SCANet model [Fig. S3(a)] demonstrates the most superior performance.

To evaluate the agronomic response of D-SCANet under varying fertilization levels, we examined LA estimations across five nitrogen (N) gradients over three sampling stages in 2023. The model captured the general upward progression of canopy expansion across growth stages and reflected the localized reduction in LA observed in the 100% N control plot, which was affected by early season flooding and stunted replanted seedlings. The estimated values followed the nonmonotonic trend of the ground-truth measurements across the evaluated treatments. The stage-specific comparison curves across all sampling dates and corresponding agronomic details are provided in Fig. S4 of the Supplementary Material.

In addition to quantitative metrics, we employed Grad-CAM [53] visualization to investigate the intrinsic decision-making mechanisms of different models during regression. Grad-CAM generates heatmaps that highlight image regions contributing most significantly to the model's final estimations. We selected four representative test samples encompassing varying plant densities, lighting conditions, and occlusion scenarios, comparing the attention patterns of our proposed D-SCANet with multiple baseline models. As illustrated in Fig. 7, the visualization results illustrate the competitive performance achieved by D-SCANet.

The first row presents a relatively sparse scenario with clearly distinguishable individual plants. In this setting, our proposed D-SCANet demonstrates precise instance-level localization capability, with its heatmap forming distinct, independent activation centers (red regions) corresponding to each water bamboo plant while completely suppressing the background water areas (blue regions). In contrast, DenseNet121 and EfficientNet exhibit overly diffuse and coarse attention patterns, generating highlight patches that cover multiple plants without effective individual discrimination. ResNet50 and VGG display fragmented and weak attention distributions, showing only minor activation points at leaf core regions, indicating their difficulty in extracting sufficient information from sparse features to make confident estimations. Meanwhile standard global attention

Models such as ViT and DeiT suffer from significant spatial diffusion into irrelevant background regions due to their lack of local inductive biases, resulting in diffuse and scattered activation. Advanced local-attention baselines like Swin-T and CAS-ViT show better spatial constraint but still fail to resolve the clear boundaries between individual plant instances as cleanly as D-SCANet.

The second row displays a complex scenario featuring uniform illumination, high leaf density, and significant interplant occlusion. On this sample, the heatmap of D-SCANet not only accurately covers all foliar regions but also reaches peak intensity in the central zone of the densest leaf overlap, demonstrating exceptional structural parsing ability. While DenseNet121 also focuses on high-density areas, its activation extends beyond leaf boundaries into background regions. Other models (e.g., EfficientNet and MobileNetV3) completely lose directional focus, distributing attention randomly across the image without forming meaningful concentration patterns.

The third-row sample was captured under clear weather conditions. D-SCANet's attention pattern remains unaffected by illumination variations, accurately localizing the core regions of all plants. Notably, standard CNNs (such as EfficientNet and DenseNet121) and advanced Transformer-based models exhibit significant misjudgments on this sample, concentrating attention on marginal or corner areas of the image while overlooking most foliar regions. This indicates that the feature extractors in these generic architectures are susceptible to being misled by illumination artifacts.

The fourth-row sample contains water surfaces densely covered with duckweeds exhibiting similar coloration to the target leaves, providing an ideal scenario to evaluate the model's feature discrimination and separation capabilities under high-frequency background noise. In this challenging environment, D-SCANet demonstrates exceptional interference resistance, with its heatmap precisely focusing on the water bamboo plants while completely suppressing surrounding duckweed areas that are shown in cold blue tones. This performance proves that our model learns a sophisticated feature extractor capable of comprehending leaf morphology, texture, and spatial structures, thereby effectively separating target plants from background noise with similar chromatic and high-frequency textural characteristics.

However, in scenarios where the water surface is densely covered by floating duckweed (see Fig. 8), nontarget vegetation exhibits chromatic and textural signatures highly similar to young crop leaves. This visual similarity introduces false-positive features into the image-level representation, accumulating background signals that lead to a slight overestimation of the physical LA. Conversely, in mature plant clusters with heavy canopy closure, leaves overlap densely along vertical stalks, creating intense mutual shading. While manual destructive sampling measures every leaf blade within the 3-D canopy volume, the top-down 2-D camera captures only the uppermost visible foliage layer. This 2-D perspective

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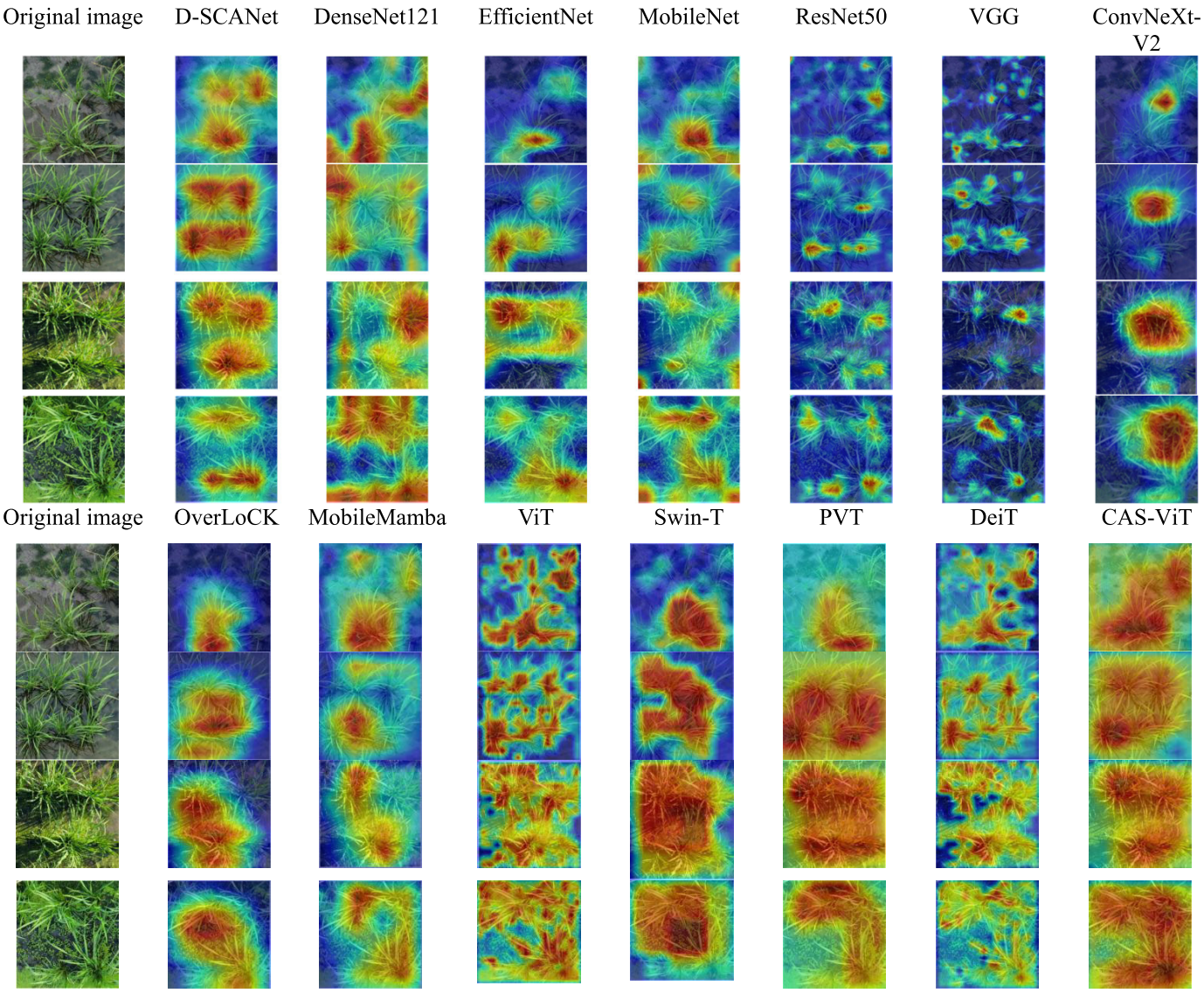


Fig. 7. Qualitative comparison of Grad-CAM visualizations across different models. The warm colors (red/yellow) indicate regions with high contribution to the final leaf area estimation, while cool colors (blue) indicate low contribution.



Fig. 8. Representative challenging field scenarios illustrating the physical limits of LA regression.



Fig. 9. MuST-C dataset.

limit conceals the lower leaf layers, causing the model to underestimate the total physical LA.

E. Cross-Site and Cross-Crop Validation

To evaluate the spatial and species transferability of D-SCANet beyond our primary training domain, we conducted a zero-shot out-of-domain evaluation using the publicly available MuST-C dataset [60]. The MuST-C benchmark (see Fig. 9) provides multitemporal UAV orthomosaic imagery and ground-truth LAI measurements across diverse field crop species, albeit with a relatively limited number of publicly available samples per crop. Evaluating our model, which was trained exclusively on aquatic water bamboo imagery in China, on this dataset under different geographic, sensor, and environmental conditions offers a challenging cross-site

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TABLE VII
ZERO-SHOT CROSS-SITE AND CROSS-CROP LAI ESTIMATION PERFORMANCE OF D-SCANET ON THE MUST-C BENCHMARK DATASET

Crop	Samples	R^2	Pearson Correlation	MAPE (%)	MAE	RMSE
Sugar Beet	4	0.75	0.87	29.04	0.72	1.13
Soy Bean	8	0.71	0.84	25.25	0.63	0.90
Wheat	16	0.21	0.46	27.14	0.28	0.36
Maize	11	0.17	0.41	43.98	1.29	1.63

and cross-crop validation. To reconcile physical resolution disparities, a dimensional scaling factor based on the ground sampling distance (GSD) ratio was applied to convert raw predicted LA (cm^2) into standardized LAI estimates.

The empirical results in Table VII demonstrate that when transferring zero-shot from water bamboo to broad-leaved crops (sugar beet and soybean), the model successfully captures general LAI variation trends, achieving R^2 values of 0.75 and 0.71, respectively. Conversely, for narrow-leaved crops (wheat and maize), the direct zero-shot validation performance shows a marked discrepancy, with R^2 values dropping below 0.22. When evaluating these narrow-leaved crops, a notable physical disparity arises from the drone flight altitudes: while our water bamboo training dataset was acquired from a low altitude of 10 m, the public dataset imagery was captured at 21 m. This higher altitude more than quadruples the physical ground footprint represented by a single pixel, causing fine leaf blades that spanned over 10 pixels in our low-altitude imagery to collapse into merely 2 or 3 pixels, smoothing out high-frequency edge gradients. This spatial coarsening is further complicated by the vertical, multitiered canopy structure of maize and wheat. As leaves overlap densely along vertical stalks, the subtle micro-shadows that indicate structural depth become optically merged within coarse pixels, making it challenging for spatial convolution layers to extract the fine geometric cues required for resolving deep canopy stacking.

V. DISCUSSION

A. Agronomic Interpretability and Architectural Advantages

At present, phenotypic detection of aquatic crops suffers from multiple water-induced interferences relative to dryland crops. Consequently, the accuracy of LA estimation, biomass inversion, and other indicators under identical models is generally lower than that for dryland crops [59]. In addition, dense tiller clumps of crops generate severe internal shading, which impairs the effective representation and modeling of leaf features. To address these challenges, our model incorporates multiple targeted optimizations. To cope with large variations in leaf dimensions across abundant foliage, gated dilated convolution is deployed for multiscale feature extraction. Furthermore, a cross-dimensional interactive attention mechanism is introduced to distinguish leaves from duckweeds, water reflections, and paddy backgrounds, so as to enhance the model's sensitivity to key leaf contours and boundaries under complex occlusions, which is crucial for achieving high-fidelity area regression.

Our comparative experiments reveal that existing general-purpose architectures that were originally designed for classification tasks fail to yield satisfactory performance when directly applied to our complex, plant-level regression task. The better performance of D-SCANet stems from its specialized coarse-to-fine information processing pipeline: DG-ASPP first provides a macro-level contextual understanding, which is then refined by SCAM's micro-level structural perception.

This architectural advantage demonstrates a more precise and semantically meaningful focus on key plant structures. The effectiveness of our approach compares favorably with related works in UAV-based LAI estimation. For instance, Wittstruck et al. [39] employed a multimodal data fusion strategy, which combined RGB imagery with geometric information from a normalized 3-D digital surface model and achieved a high R^2 of 0.83 on winter wheat LAI estimation in field level. In another study, Ilniyaz et al. [1] reported a standard ResNet architecture used for LAI prediction in vineyards with an R^2 of 0.82. In contrast, our D-SCANet used only RGB images and obtained plant-level LA values. The numerical difference in R^2 between our method and multimodal approaches like that of Wittstruck et al. [39] highlights the significant informational value provided by explicit 3-D data such as nDSM. However, our model's performance surpassed SOTA architectures, which reflects that the integration of DG-ASPP and SCAM modules leads to a more effective use of information and provides a viable alternative for scenarios limited to RGB sensors.

B. Model Limitations and Future Work

The primary limitation of D-SCANet is that the model's predictive accuracy remains suboptimal. This is primarily because the RGB images utilized by the model only capture the visible plant canopy, omitting internal or occluded leaves, whereas manual measurements calculate the LA of each individual leaf. Despite this inherent bias, the true and estimated values demonstrate a good fit, implying that estimated values can be effectively corrected to approximate true LA through a linear transformation. Regarding deployment efficiency, the parameter footprint of D-SCANet (48.20M) remains relatively high for resource-constrained edge platforms, such as low-power field microcontrollers or lightweight drones. Furthermore, the evaluation of cross-species generalizability is constrained by the scope and imaging conditions of available benchmarks. As observed in our cross-crop evaluation (see Table VII), while the model demonstrates promising zero-shot transferability to broad-leaved crops like sugar beet (R^2 = 0.75) and soybean (R^2 = 0.71), its metrics on narrow-leaved row crops such as maize (R^2 = 0.17 and MAPE = 43.98%) and wheat (R^2 = 0.21) remain constrained. This discrepancy primarily reflects the limited sample size of the public benchmark and the mismatched drone shooting altitudes, which altered the spatial pixel resolution of fine leaf blades. Therefore, future work will focus on systematically validating the application performance of D-SCANet on narrow-leaved crops under con-

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sistent shooting flight altitudes, matched spatial resolutions, and expanded multicrop datasets.

To break through the current performance and expand model universality, future research will advance along three synergistic dimensions: dataset expansion, training paradigms, and modality strategies.

Building upon these identified boundaries, future research will advance toward a more universal and high-precision plant phenotyping framework. To expand dataset scale without overwhelming manual labor, we plan to deploy active learning and semi-supervised sampling during young seedling stages to build efficient ground-truth baselines, while progressively expanding our field data collection across diverse growth stages, geographical sites, and crop species. Architecturally, breaking the performance ceiling of single-modality observations requires extending the model's physical representation beyond 2-D RGB imagery. Future efforts will integrate canopy structural priors, such as leaf inclination angle, gap fraction, and multilayer shading, while fusing multispectral, textural, and multitemporal phenological data.

Furthermore, rather than training models from scratch for novel crop species, we will explore parameter-efficient fine-tuning (PEFT) techniques, such as adapters or low-rank adaptation (LoRA), applied directly to D-SCANet. By freezing the pretrained feature extractor and fine-tuning only a minimal fraction of adapter parameters on small target-crop samples to achieve rapid, low-cost cross-crop adaptation. Concurrently, leveraging emerging VLMs in conjunction with multimodal remote sensing data offers a compelling avenue toward developing intelligent, multimodal quantitative agricultural vision systems. Finally, knowledge distillation and network pruning will be investigated to compress these refined architectures into ultralightweight student models, thereby bridging the gap between high-performance research models and real-time edge-device deployment in field agriculture.

Although the LA estimation accuracy of our model has not yet reached the optimal level, it can acquire plant-scale LA values when exhaustive manual ground-truth sampling of every individual plant across the entire field plot is infeasible. With the accumulation of manually measured datasets and continuous improvements to the model, we expect estimation accuracy to be further improved in the near future.

VI. Conclusion

This study addresses the challenging task of estimating multiplant canopy LA from UAV RGB images in complex field environments. We proposed an innovative regression model D-SCANet, which cascades two custom modules: the DG-ASPP for adaptive multiscale context aggregation, and the SCAM that precisely captures fine leaf spatial information through cross-dimensional interactions. Extensive experiments on our self-built water bamboo dataset demonstrate D-SCANet's superiority, outperforming various mainstream networks across all key metrics with a minimum MAPE of 20.34% and maximum R^2 of 0.78. This study explores and validates a promising marker-free approach for high-throughput plant phenotyping analysis. The design of D-SCANet also provides references for complex visual tasks in other crops with slender and dense foliage.

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